

Taewoo Kim

EIT (eligible June 2026)

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Education

B.Eng., Electrical Engineering, The British Columbia Institute of Technology

January 2022-2026

GPA (Cumulative): 92% (4.00/4.00)

Selected relevant coursework: Power System Components (100%) | Electrical Power System Analysis & Design (88%) | Electromagnetic Fundamentals (100%) | Advanced Circuit Analysis (90%) | Industrial Control Systems (91%) | Signal Processing and Filters (99%) | Microcontroller Systems (96%) |

Research Interests: Grid modernization | Smart Grid | Renewable Energy Resources | Power system stability | Power system modelling, and planning | DER and EV integration |

Work Experience

Substation Intern, Part-time, Xentric Engineering (Langley, B.C)

Sep 2025 – June 2026

- Supported the design and review of substation physical layouts and elevation drawings for utility projects.
- Reviewed vendor drawings, datasheets, and technical specifications to extract equipment information for engineering design and procurement support.
- Drafted substation single-line & three-line diagrams in AutoCAD for electrical design deliverables.
- Configured RTAC & built HMI using SEL-5033 & SEL-5035, including IED communication setup, RTAC tag mapping, SCADA point integration, and IEC 61131-3-based automation logic exposure.

OH Distribution Designer, Full-time & Part-time, Actalent (New Westminster, B.C)

Feb 2025 – Sep 2025

- Worked on BCH overhead distribution pole maintenance designs using ES43 standards
- Worked on a telecom preliminary project involving pole modeling using SPIDAcad & PLS-CADD, verifying structural strength and clearances in accordance with ES43 standards
- Engaged in collaborative with cross-functional teams, such as engineers, technologists, city and municipalities, and customers effectively improved my communication skills

Electrical Engineering Student, Co-op, BCHydro (Edmonds, B.C)

May 2024 – Aug 2024

- Familiar with transmission components and structures, contributing to inspection and design processes
- Prepared a proposal document for the replacement of a transmission circuit line due to defects
- Assisted engineers with substation system map design mark-up, contributing to precise and effective designs
- Conducted inspections in PG RAU, ensuring high-quality inventory information by cross-referencing TLDS data
- Developed a VBA tool for TAR forms, streamlining communication between field operators and the team

Design Assistant, Co-op & Part-time, ZE Power Engineering (Richmond, B.C)

Jan 2023 – May 2024 &
Aug 2024 – Feb 2025

- Worked on BCH underground feeder cable replacement designs using ES53 standards
- Worked on BCH overhead distribution pole maintenance designs using ES43 standards

Capstone Project

Utility Health Monitoring System, BCIT School of Energy

Sep 2025 – May 2026

Supervisor: Prof. Ed Casas

Designed and implemented a proof-of-concept utility pole monitoring system that measures pole tilt and moisture using ESP32-C3 sensing nodes, an LIS3DH accelerometer, and an FDC1004 capacitance sensor. The system transmitted data through an ESP-based wireless mesh network to a fixed root node, then forwarded readings to a PC application for SQLite database storage, dashboard visualization, and offline map-based monitoring. The project integrated embedded firmware, PCB design, sensor calibration, wireless communication, and enclosure design for a battery-powered pole-mounted prototype.

Projects

Stream Deck HID (RPi Pico 2W & FreeRTOS)

Oct – Dec 2025

- Co-developed a real-time Stream Deck HID device using Raspberry Pi Pico 2W, capacitive touch buttons, a rotary encoder and an SSD1306 OLED display using C
- Implemented FreeRTOS-based firmware with separate tasks for BLE communication, GPIO setup, OLED display updates, input monitoring, and encoder event processing

- Designed an interrupt driven BLE event pipeline using GPIO ISRs, FreeRTOS queues, and a Python Bleak host script to trigger customizable responses from hardware inputs

Optical Satellite Simulation (DE0-Nano SoC & HS-SR04)

Feb – Apr 2025

- Designed an FPGA based optical satellite communication simulator using DE0-Nano-SoC, Verilog, laser-based OOK modulation, and UART communication
- Implemented data encoding and decoding logic to transmit text through optical on and off pulses and reconstruct the message on the receiver side using system verilog
- Built a PC interface using UART to display decoded text and allow configurable transmission frequency

Autonomous Rover (RPi & OpenCV)

Feb – Apr 2025

- Developed a Raspberry Pi 4 based rover in C++ using object-oriented programming, with modular classes for vehicle control, Aruco marker detection, TCP client/server communication
- Implemented dual-mode operation, including controller teleoperation with live video streaming and OpenCV based autonomous navigation using marker detection, camera calibration, and distance estimation
- Designed mechanical and electrical integration, including a SolidWorks 3D model, a custom RPi Hat PCB in KiCad

Autonomous Ball-Sorting Machine (RPi & OpenCV)

Feb – Apr 2025

- Developed a Raspberry Pi based autonomous ball sorting machine using C++ OOP and OpenCV to detect object colors and classify materials
- Integrated a simple sorting system using a motor and MCP3008 ADC to control object placement based on image processing results

BC One Call automation (Python)

Feb – Apr 2024

- Developed an automation tool using python with user-friendly interface to automate data extraction, processing, and input preparation for BC1C ticket submissions, improving workflow efficiency and consistency

Portable DC Power Supply (Altium)

Feb – Apr 2023

- Designed a portable DC power supply PCB in Altium, including schematic, component placement, and board layout
- Created 2D and 3D technical drawings of the power supply chassis using AutoCAD to support mechanical assembly
- Assembled and tested the PCB by hand soldering both through hole and SMD components

Competitions | Scholarship | Award

Western Engineering Competition (University of Saskatchewan)

Jan 2025

Competed in Programming, developed a competition solution for a real-world urban planning challenge focused on optimizing fire hall placement as cities expand. Our approach aimed to reduce long-term community costs and future infrastructure impacts by using MATLAB for backend optimization/modeling and Python to present the results through a web-based interface. The project repository shows development of GUI, database, and API components for the Western Engineering Competition, with MATLAB and Python as the main implementation languages.

BCIT Engineering Competition – Senior Design

Second Place

Won 2nd place at BCIT's Senior Design BEC Competition, where our team designed and built a wirelessly controlled rescue vehicle within a two-hour time limit. The vehicle used an Arduino, motors, and sensors to detect an "injured pedestrian," navigate the course, and safely transport them to a designated hospital location. The project demonstrated rapid prototyping, embedded systems design, sensor integration, motor control, and team-based engineering problem-solving under time constraints.

BCIT The Robert B. Wells and Family Award

Oct 2024

BCIT International Student Award

Oct 2023

- Achieved highest GPA among international in BCIT

BCIT Faculty and Staff Association Scholarship

Oct 2023

Technical Skills

Programming & Tools:	C, C++ Python MATLAB/Simulink System Verilog OpenCV VS Code VS Studio SolidWorks ETAP OpenDSS PLS-CADD
Embedded & Interfaces:	TI MCUs Raspberry Pi RPi Pico 2W Arduino ESP32 UART/SPI/I2C Firmware development
Hardware & PCB:	KiCad & Altium PCB layout soldering & assembly 3D printing
Test & Measurement:	Oscilloscope DMM Logic Analyzer Spectrum Analyzer

Research Interest

Topic

Quantifying the Modeling Adequacy of Static and Dynamic Load Models for Hyperscale Data Center Interconnection Studies

Research Objective

The main goal of this research is to evaluate when static models are sufficient for large data center interconnection studies and when dynamic load models are necessary. The study will compare constant-P/Q, ZIP, and dynamic data center load models under voltage disturbance scenarios, with particular focus on how grid strength, short circuit ratio (SCR), affects model divergence.

Three Research questions

1. How do constant-P/Q, ZIP, and dynamic data center load models differ in predicting transient voltage recovery after grid faults?
2. How does grid strength, SCR, affect the divergence between static and dynamic load model results? At what SCR range does the static model become insufficient?
3. Can a practical model selection guide be developed to support a planner's decision?

Short Methodology

The research will use a simulation-based approach on a simplified transmission and sub-transmission test system with a large data center load connected at a point of interconnection. The study would compare constant-P/Q, ZIP, and dynamic load models under selected voltage-disturbance scenarios.

Grid strength would be varied by changing the equivalent source impedance or short circuit capacity at the point of interconnection. The main outputs would include voltage nadir, voltage recovery time, load disconnection amount, reactive power response, and a model divergence index comparing static and dynamic simulation results.

Intended Contribution

My intended contribution is not to develop a new data center dynamic model, but to quantify the conditions when a static model becomes inadequate for large data centre interconnection studies.